



GOSFORD HIGH SCHOOL

Year 11

2010

Preliminary HSC MATHEMATICS

Assessment Task 1

Time Allowed: 65 minutes

Instructions:

- Remember to start each new question on a new page.
- Students must answer questions using a blue/black pen and/or a sharpened B or HB pencil.
- Approved scientific calculators may be used.
- Students need to be aware that:
 - All necessary work must be shown in order to gain full marks.
 - Untidy and/or poorly organised solutions may not gain full marks.

QUESTION 1: (14 Marks)

		Marks
1.	Find the value of	
a.	$\frac{4.5 \times 10^6}{9.8 \times 10^{-5}}$ To 3 significant figures.	1
b.	$\frac{5\left(\left(\frac{2}{3}\right)^5 - 1\right)}{\sqrt[5]{5}}$ Answer in scientific notation correct to 2 significant figures.	2
2.	An antique vase is bought and sold for \$1120 making a profit of 27 ½%. What was the original cost price?	2
3.	Express $2.1\dot{3}\dot{6}$ as a mixed fraction.	2
4.	Convert 300 metres in 4 seconds to km/hr.	1
5.	Express each of the following in simplest surd form.	
i.	$\sqrt{675}$	
ii.	$(2\sqrt{2} - \sqrt{6})(2\sqrt{3} - 1)$	3
6.	Express $\frac{\sqrt{3}}{2\sqrt{3}+1}$ with a rational denominator	2
7.	Evaluate exactly. (Express without absolute value symbols.)	
i.	$ \sqrt{3} - \sqrt{2} $	
ii.	$ 2\sqrt{3} - \sqrt{13} $	1

QUESTION 2: (19 Marks)

	Marks
1. Expand and simplify	
$(3m - 1)(2m^2 + m + 3)$	2
2. Simplify	
$\frac{y}{x^2 + xy} - \frac{1}{x}$	2
3. Show that $x = 2\sqrt{2} - 3$ is a solution to the equation $x^2 + 6x + 1$	2
4. Write the following in index notation and without a denominator	2
i. $\frac{1}{3x+7}$ ii. $\frac{4}{(\sqrt{x-2})^3}$	
5. Fully factorise	
i. $(m + n)^2 + 5(m + n)$	2
ii. $x^3 + 5x^2 - 9x - 45$	2
iii. $x^3 - 8$	2
iv. $26x^2 + 13x - 13$	2
v. $3a^4 - 3$	3

QUESTION 3: (14 Marks)

	Marks
1. Solve exactly	
i. $3x - 5 = 7x + 8$	1
ii. $\sqrt{5x} - 2 = 3 - x$	2
iii. $5 - 4x > -8$	2
iv. $5 = \frac{x}{2} - \frac{4}{x}$	2
2. Solve simultaneously	
$y = \frac{1}{x}$ and $2x + 3y - 7 = 0$	3
3. Solve	
$9^{7x-2} = \left(\frac{1}{27}\right)^{-x}$	2
4. Solve	
$ 4 - 3x \leq 13$	2

QUESTION 4: (17 Marks)

			Marks
1.	i.	Sketch the parabola $y = x^2 + 6x + 14$ (Clearly labelling the coordinates of the vertex)	2
	ii.	Hence find the minimum value of $x^2 + 6x + 14$	1
	iii.	Hence determine the number of solutions to $x^2 + 6x + 14 = 0$	1
2.	Complete the square hence find the maximum value of $y = 6 - x - x^2$		3
3.	Solve		
	i.	$(x - 3)(x + 4) \geq 0$	2
	ii.	$-5x^2 + 4x - 1 < 0$	2
4.	Find the value(s) of p such that $x^2 + (p + 1)x + 4 = 0$ has real and different roots.		3
5.	Find the value(s) for p for which the equation $x^2 - 2px + 8p - 15 = 0$ has two equal roots (ie only one solution)		3

END OF TEST

Solutions Assessment 1. Preliminary

Question 1.

- 1 (a) 4.59 (b) -1.2×10^0
- 2 $C \times 1.275 = 1120$
 $C = 8828.43$
- 3 $2 + 1.3\bar{6}$
 $x = 1.3\bar{6}$
 $100x = 136.\bar{6}$
 $x = \frac{136.\bar{6}}{99} = \frac{3}{22}$
 Ans $2\frac{3}{22}$
- 4 $300 \text{ m in } 4 \text{ sec} = 75 \text{ m/s}$
 $= \frac{75 \times 60 \times 60}{1000}$
 $= 270 \text{ km/hr}$
- 5 (i) $15\sqrt{3}$
- (ii) $4\sqrt{3} - 2\sqrt{2} - 2\sqrt{18} + \sqrt{6}$
 $4\sqrt{3} - 2\sqrt{2} - 6\sqrt{2} + \sqrt{6}$
 $5\sqrt{6} - 8\sqrt{2}$
- 6 $\frac{\sqrt{3}}{2\sqrt{3}+1} \cdot \frac{(2\sqrt{3}-1)}{(2\sqrt{3}-1)} = \frac{6-\sqrt{3}}{11}$
- 7 (i) $\sqrt{3} - \sqrt{2}$ (ii) $\sqrt{3} - 2\sqrt{3}$

Question 2

- ① $6m^3 + 3n^2 + 9m - 2m^2 - n - 3$
 $6m^3 + n^2 + 9m - 3$
- ② $\frac{y}{x^2+xy} - \frac{1}{x}$
 $\frac{y}{x(x+y)} - \frac{(x+y)}{x(x+y)}$
 $\frac{-x}{x(x+y)} = \frac{-1}{x+y}$
- ③ sub $x = 2\sqrt{2} - 3$ into $x^2 + 6x + 1 = 0$
 $(2\sqrt{2} - 3)^2 + 6(2\sqrt{2} - 3) + 1 = 0$
 $8 - 12\sqrt{2} + 9 + 12\sqrt{2} - 18 - 3 = 0$
 $0 = 0$
- ④ (i) $(3x+1)^{-1}$ (ii) $4(x-2)^{-\frac{3}{2}}$
- ⑤ (i) $(m+n)(m+n+5)$
- (ii) $x^3 + 5x^2 - 9x - 45$
 $x^2(x+5) - 9(x+5)$
 $(x^2-9)(x+5)$
 $(x-3)(x+3)(x+5)$
- (iii) $(x-2)(x^2+2x+4)$
- (iv) $26x^2 + 13x - 13$
 $13(2x^2 + x - 1)$
 $13(2x-1)(x+1)$

5 (a) $3a^4 - 3$
 $3(a^4 - 1)$
 $3(a^2 - 1)(a^2 + 1)$
 $3(a-1)(a+1)(a^2 + 1)$

Question 3

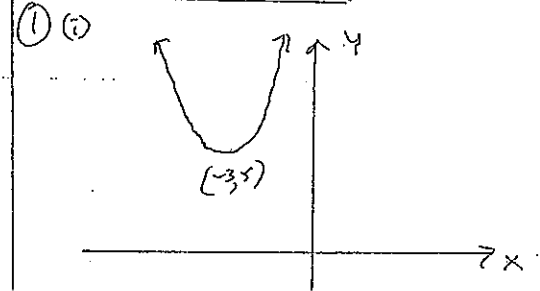
- ① (i) $3x - 5 = 7x + 8$
 $x = -\frac{13}{4}$
- (ii) $\sqrt{5}x - 2 = 3 - x$
 $\sqrt{5}x + x = 5$
 $x(\sqrt{5} + 1) = 5$
 $x = \frac{5}{\sqrt{5} + 1}$ or $\frac{5\sqrt{5}-5}{4}$
- (iii) $5 - 4x > -8$
 $-4x > -13$
 $x < \frac{13}{4}$
- (iv) $5 = \frac{x}{2} - \frac{4}{x}$
 $10x = x^2 - 8$
 $x^2 - 10x - 8 = 0$
 $x = \frac{10 \pm \sqrt{100 + 32}}{2}$
 $= \frac{2(5 \pm \sqrt{32})}{2}$
 $= 5 \pm \sqrt{32}$

② $y = \frac{1}{2}$ $2x + 3y - 7 = 0$
 sub $y = \frac{1}{2}$ $2x + \frac{3}{2} - 7 = 0$
 $2x^2 - 7x + 3 = 0$
 $(2x-1)(x-3) = 0$
 $x = \frac{1}{2}$ or $x = 3$ $y = \frac{1}{2}$
 $(\frac{1}{2}, \frac{1}{2})$ or $(3, \frac{1}{2})$

③ $9^{2x-2} = (\frac{1}{27})^{-x}$
 $(3^2)^{2x-2} = (3^{-3})^{-x}$
 $14x - 4 = 3x$
 $11x = 4$
 $x = \frac{4}{11}$

④ $|4 - 3x| \leq 13$
 $4 - 3x \leq 13$ or $4 - 3x \geq -13$
 $-3x \leq 9$ $-3x \geq -17$
 $x \geq -3$ $x \leq \frac{17}{3}$

Question 4



$y = x^2 + 6x + 14$
 Axis of symmetry $x = -\frac{b}{2a}$
 $x = -\frac{6}{2}$
 $x = -3 \Rightarrow y = 5$

(ii) Minimum value 5

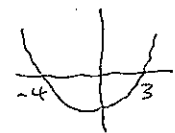
(iii) No Solutions
 Does not cut x axis

$$\begin{aligned}
 ② \quad y &= 6 - x - x^2 \\
 &= -(x^2 + x - 6) \\
 &= -(x^2 + x + (\frac{1}{2})^2 - \frac{1}{4} - 6) \\
 &= -[(x + \frac{1}{2})^2 - 6\frac{1}{4}] \\
 &= -(x + \frac{1}{2})^2 + 6\frac{1}{4}
 \end{aligned}$$

$\therefore$ max value $6\frac{1}{4}$ at $x = -\frac{1}{2}$

$$③ \text{ (i)} \quad (x-3)(x+4) \geq 0$$

consider $x=3$ $x=-4$



From sketch

$$x \geq 3 \text{ or } x \leq -4$$

OR Test $x=0$

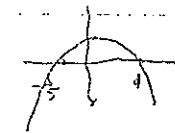
$$-3 \times 4 \geq 0 \text{ False}$$

$$\therefore x \leq -4 \text{ or } x \geq 3$$

$$③ \text{ (ii)} \quad -5x^2 + 4x + 1 \leq 0$$

$$(5x+1)(-x+1)$$

consider $x = -\frac{1}{5}$ or 1



From sketch

$$x \leq -\frac{1}{5} \text{ or } x \geq 1$$

OR Test $x=0$

$$1 \leq 0 \text{ False}$$

$$\therefore x \leq -\frac{1}{5} \text{ or } x \geq 1$$

$$④ \quad x^2 + (p+1)x + 4 = 0$$

for real and different roots

$$\Delta > 0$$

$$\text{i.e. } (p+1)^2 - 16 > 0$$

$$p^2 + 2p + 1 - 16 > 0$$

$$p^2 + 2p - 15 > 0$$

$$(p+5)(p-3) > 0$$

$$p < -5 \text{ or } p > 3$$

$$⑤ \quad x^2 - 2px + 8p - 15 = 0$$

for two equal roots $\Delta = 0$

$$\text{i.e. } (-2p)^2 - 4(8p - 15) = 0$$

$$4p^2 - 32p + 60 = 0$$

$$4(p^2 - 8p + 15) = 0$$

$$4(p-3)(p-5) = 0$$

$$p=3 \text{ or } p=5$$

Corrections

Question 2 ~~part 3~~

part 3

show that $x = 2\sqrt{2} - 3$ is a solution to the equation $x^2 + 6x + 1 = 0$

Question 4

part 3(ii)

$$-5x^2 + 4x + 1 < 0$$